Multiple Linear Regression

1. What You'll Learn in This Section

You'll learn to:

- Explain how multiple linear regression combines several predictors to estimate one output.
- Select and inspect features based on the dataset and prediction problem.
- Build, train, and test a multiple linear regression model with scikit-learn.
- Compare models using MAE, MSE, RMSE, and R-squared.

2. Detailed Explanation

From one predictor to many

Linear regression learns a linear equation that connects known inputs to an unknown output. Inputs are also called features, independent variables, or predictors. The dependent variable is written as y, while a prediction is written as $\hat{y}$.

Simple linear regression uses one predictor:

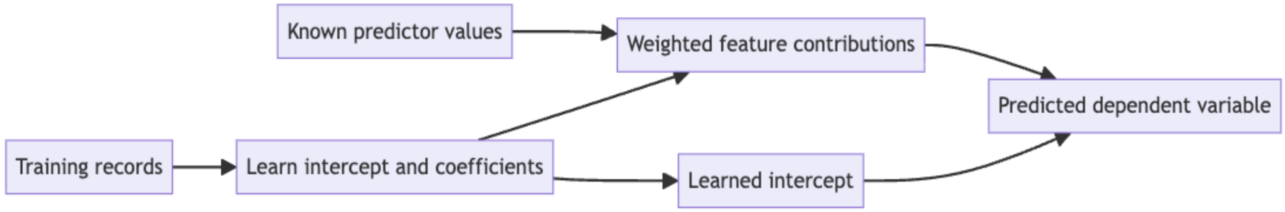
$$\hat{y} = w_0 + w_1x_1$$

The intercept w_0, or bias, is the prediction when the feature is zero. The coefficient w_1 is the change in the prediction for a one-unit increase in x_1.

Multiple linear regression keeps the same idea but uses two or more predictors:

$$\hat{y} = w_0 + w_1x_1 + w_2x_2 + \cdots + w_nx_n$$

Each feature supplies a separate, weighted contribution. The model still predicts one dependent variable. For salary, the predictors might include experience, education, and communication skill, while salary remains the single output.



Relevant features and representative instances

A model can receive more useful information in two complementary ways:

- Add relevant features.
- Add representative instances, examples, or data points.

An employee record is one instance. Experience, education, and communication skill are possible features in that record. More instances help represent the population, while more relevant features describe each instance in greater detail.

The collection cost depends on the problem. A company may already hold many employee or house-price records. Gathering detailed skill scores may require forms and manual work. In biomedical work, even collecting enough instances can be difficult.

Random or irrelevant features do not provide the benefit of relevant information. More features also increase computation and collection effort, so usefulness must be confirmed experimentally.

Feature relevance depends on the prediction setting

A feature does not have one universal level of importance. Its value depends on the role, population, encoding, other predictors, and training records.

Consider three prediction settings:

Prediction	Possible predictors	Dataset-specific concern
Salary	Experience, education, communication, leadership, certifications, skills, and performance	Different roles reward different qualifications.
House price	Bedrooms, age, size, amenities, and location	Age may lower value, while size or amenities may raise it.

Final examination performance	Study hours, attendance, and previous GPA	Attendance and previous GPA may not describe every student equally well.

Scores can encode attributes such as communication or programming ability. The score must remain consistent with the performance it represents. Detailed features can be valuable, but collecting them can be expensive.

Choosing between simple and multiple models

More predictors may improve accuracy, but a simpler model is preferable when it performs as well as a more complex one.

Suppose houses in one confined area share nearly the same distance from the city centre, amenities, and bedroom count. House size may be the only feature that varies enough to explain price. The comparison is then practical:

1. Train a simple model using size alone.
2. Train a multiple model using size and the other available features.
3. Evaluate both models on the same problem.
4. Keep the simpler model if their performance is approximately the same.

The simpler system can ask a customer only for house size. This reduces collection effort and complexity. Neither model universally replaces the other; the decision comes from experimental comparison.

Inspecting features with scatter plots

Simple linear regression fits naturally on a two-dimensional plot: one feature on the x-axis and the output on the y-axis. Multiple predictors and one output do not fit together on one ordinary two-dimensional plot.

Instead, plot the output separately against each feature. For a salary dataset, these plots may reveal:

Experience follows a relationship close to a straight line and may be a strong predictor.
Education levels contain different salaries because other features also contribute.
Age shows a trend but may repeat information already supplied by experience.
A curved or highly irregular pattern is a poor match for a straight-line model.

To test redundancy, train one model with age and another without it. If performance changes insignificantly, age may be reproducing experience information. Record order can make a table easier to read, but it does not change the relationships shown in a scatter plot.

Watch out for: A visible trend does not prove that a feature adds unique value. Compare model performance with and without the feature.

Reading the intercept and coefficients

The algorithm learns the intercept and coefficients from training data by minimizing its selected training error criterion. For ordinary least squares, this criterion is the sum of squared residuals.

Consider this salary equation:

[\hat{y} = 25{,}000 + 4{,}000x_1 + 3{,}000x_2 + 500x_3]

Here, ₹25,000 is the base salary. Each year of experience adds ₹4,000, each encoded education level adds ₹3,000, and a binary communication value of 1 adds ₹500. A value of 0 adds nothing for that communication term.

A positive weight raises the prediction as its feature increases. A negative weight lowers it. Coefficient magnitudes are directly comparable only when the features use comparable scales. The effect on one prediction also depends on the encoded feature value.

Learned coefficients do not have to stay between 0 and 1. Do not force them into a preferred range. Improve missing features or insufficient data instead of manually changing learned weights.

Preparing and splitting data with scikit-learn

The workflow uses pandas for tabular data, NumPy for numerical work, Matplotlib for plotting, and scikit-learn for splitting, fitting, predicting, and evaluation.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
```

Inspect the table before modelling. Check that the expected columns exist, then look for missing values, implausible encodings, and relationships poorly matched by a linear model.

Machine learning code must separate predictors from the target explicitly. A displayed row index is not an input feature. In the salary example, `X` contains Experience, Education, and Age, while `y` contains Salary.

This split reserves 20% of the records for testing and 80% for training. A fixed `random_state` repeats the same split. Preserve the returned order: `X_train`, `X_test`, `y_train`, and `y_test`.

A 70/30 experiment can use `test_size=0.3`. Reserve validation data only when a validation task is actually present.

Fitting, predicting, and inspecting the model

`LinearRegression()` creates the model object. Its `fit` method learns from `X_train` and `y_train`. Its `predict` method estimates `y_pred` from `X_test`.

A new candidate record must use the same feature order used during fitting. A pandas DataFrame with matching column names helps preserve that order.

The predictions in `y_pred` are outputs. The intercept and coefficients are learned weight parameters. Libraries are generally preferred to hand-written implementations because their routines are optimized and repeatedly improved, but the code must still follow the library's conventions.

Evaluating prediction quality

Evaluation compares actual test values, `y_test`, with predictions, `y_pred`. Smaller errors and a higher test-set R-squared indicate a better fit when models use the same data.

Metric	What it measures	Preferred direction
MAE	Average absolute difference between actual and predicted values	Lower
MSE	Average squared difference between actual and predicted values	Lower
RMSE	Square root of MSE	Lower
R-squared	Goodness of fit between the learned function and actual data	Higher, closer to 1

MAE and RMSE use the target's original unit. MSE uses squared target units, so its numerical magnitude should not be compared directly with RMSE. Squaring gives larger errors more influence, so RMSE is at least as large as MAE for the same predictions.

Training error confirms that the model fits data it has seen. Reported performance should use the unseen test set because that better represents use outside development. Acceptable performance depends on the metric, data, and cost of errors.

Limits and boundary cases

Linear regression expects the output to be approximately a linear function of its inputs. Curved, clustered, or highly irregular relationships may not fit a straight line well. The model can still serve as a baseline, but poor performance may require another algorithm.

Outliers are exceptional points that can pull the fitted line away from ordinary records. Investigate them instead of removing them automatically:

- Correct or remove incorrect records.
- Keep legitimate exceptional cases visible during evaluation.
- Add directly relevant features when they can explain a valid exceptional outcome.

The formulation here predicts one dependent variable from multiple predictors. Problems with multiple discrete outcomes belong to classification. Positive, neutral, and negative sentiment are classes, as are the digits 0 through 9.

3. Key Takeaways

- Multiple linear regression adds weighted contributions from several predictors to estimate one dependent variable.
- Relevant features and representative instances can help, but feature value must be tested on the selected dataset.
- Scatter plots and model comparisons reveal linear patterns, redundant features, and unnecessary complexity.
- Train on one subset, evaluate on unseen test data, and interpret metrics in their proper units.
- Nonlinear patterns and outliers can limit the quality of a fitted linear model.

Mental model: Think of multiple linear regression as an expanded (y = mx + c), where each predictor adds its own learned weighted contribution.